Task-Oriented Semantic Coding and Utility Optimal Transmission in Satellite-Integrated Internet

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***Abstract*— The upcoming satellite-integrated Internet can provide onboard remote sensing image processing and efficient communication to ensure ubiquitous intelligent services. Given the massive volume of remote sensing images, the efficient extraction and transmission of task-oriented information to the corresponding user equipment (UE) remains a critical challenge. To address this challenge, we propose a task-oriented semantic coding and utility-optimal transmission (TUT) framework for satellite-integrated Internet. Specifically, we propose a metric named utility loss of information (UoI) to simultaneously capture the freshness, task updates, and task completion of UEs. Building upon this metric, our TUT framework leverages perceptual-weight maps (PMs) generated from the remote sensing images which allow for variable code rates specific to the tasks of UEs. Besides, the TUT framework can dynamically adjust the numerical distribution of PM to optimize semantic coding tailored to UoI. Considering limited onboard resources, we further model a long-term UoI minimization problem by utilizing the Lyapunov optimization framework, decomposing it into a set of single-slot problems, and employing a proximal policy optimization (PPO) algorithm to solve this nonconvex UoI minimization problem. Simulation results demonstrate that our TUT framework can achieve minimum long-term average UoI and power consumption compared to the state-of-the-art schemes.**

***Index Terms*— Remote sensing image, resource allocation, satellite-integrated Internet, task-oriented semantic coding, utility loss of information (UoI).**

NOMENCLATURE

S	Semantic-empowered satellite.
U	Set of K user equipment per beam.
T	Number of time slots in the transmission period.
p_r	Probability of remaining in the current task state.
p_t	Probability of transitioning to another task state.
$\boldsymbol{x}_i(t)$	Image corresponding to UE_i at time slot t.
$\boldsymbol{q}_i(t)$	PM corresponding to $\boldsymbol{x}_i(t)$ at time slot t.
$\boldsymbol{y}_i(t)$	Semantic information extracted from $\boldsymbol{x}_i(t)$.
$\mathbf{p}_i(t)$	Transmit power vector for UE_i.
$\mathbf{s}(t)$	Superimposed signal.
$\boldsymbol{r}_i(t)$	Signal received by UE_i.
$R_i(t)$	Data rate of UE_i.
$\hat{\boldsymbol{y}}_i(t)$	Signal separated by SIC.
$\hat{\boldsymbol{x}}_i(t)$	Reconstructed image.
$D_i(t)$	Current state of the source.
$\hat{D}_i(t)$	Perceived task state of UE_i.
$\Delta_i(t)$	Utility loss of information for UE_i.
$e(\cdot)$	Semantic encoder in the PMSC network.
$d(\cdot)$	Semantic decoder in the PMSC network
$P_{\max}$	Maximum total power.
P_{mean}	Long-term average power.
$Q_i(t)$	Data packets waiting to be transmitted to UE_i.
$A_i(t)$	Arrive data rate determined by the volume of $\boldsymbol{y}_i(t)$.
$P(t)$	Power debt queue.
$V(t)$	Virtual queue state at time slot t.
$L(V(t))$	Quadratic Lyapunov function.
$\text{LD}(V(t))$	Lyapunov drift.
$\text{LP}(V(t))$	Lyapunov penalty term related to UoI.
$\text{DPP}(V(t))$	DPP term of the TUT framework.
v	Weight of the penalty term, $v \geq 0$.

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I. INTRODUCTION

SATELLITE-INTEGRATED Internet is regarded as a future information infrastructure that supports remote sensing, communication, and other ubiquitous intelligent services, which is also the core component of the sixth-generation (6G) network [1], [2]. In applications of satellite-integrated Internet with remote sensing image transmission, such as disaster relief, aviation, and navigation monitoring, the efficient transmission of images according to the task requirements of ground user equipment (UE) is crucial [3]. However, the volume of image data generated by remote sensing satellites reaches terabyte (TB) level per day [4], exacerbating the conflict between the massive images generated onboard and the limited transmission capacity of the satellite-to-terrestrial

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link. This conflict poses significant challenges in meeting the demands for timeliness and accurate information acquisition by UEs [5].

With the development of high-resolution remote sensing imagery, the low compression ratio (CR) of conventional image source coding schemes based on discrete cosine transform (DCT) [6] or discrete wavelet transform (DWT) [7] no longer satisfy the processing and storage for the massive volume of remote sensing images [8], [9]. Moreover, the independent structures of conventional source and channel coding also exhibit a "cliff effect" in that the system performance experiences a sharp decline when the channel signal-to-noise ratio (SNR) falls below a critical threshold [10]. Semantic coding, which adopts a joint source-channel coding (JSCC) structure, can efficiently extract and transmit task-oriented information according to the task requirements of UEs, sustaining commendable performance with inferior channel conditions at high CR [11], [12]. Consequently, considering the conflict of satellite-integrated Internet, semantic coding cooperated with valid resource allocation transmission policy presents a potential solution for ensuring the freshness and accuracy of transmitting the massive volume of remote sensing images [13], [14].

A. Related Works

1) Image Semantic Transmission System: Many existing image semantic transmission systems primitively focus on optimizing data-oriented metrics, like peak SNR (PSNR) or multiscale structural similarity (MS-SSIM) at specific code rates, which are not tailored to specific tasks [15], [16]. Subsequently, leveraging the task-aware capability of semantic coding, some studies begin to incorporate task-oriented features into the optimization metric. For instance, Cheng et al. [17], [18] examined the issues of semantic coding and resource allocation for image transmission to optimize quality of experience (QoE), which considers the date rate, bit error probability (BER), and interest rating.

Although QoE integrates latency and information value metrics, it does not explicitly quantify the temporal freshness of task updates at the receiver. To address this gap, age of information (AoI) [19], defined as the time elapsed since the latest information generation, has been adopted to measure freshness in semantic communication systems [20], [21], [22], [23]. However, AoI focuses on temporal aspects and neglects the contextual or task-specific value of information (VoI). Emerging metrics such as VoI [24], age of incorrect information (AoII) [25], and just noticeable difference (JND) [26] aim to bridge this gap by jointly assessing freshness and task relevance. For instance, AoII evaluates both dimensions by quantifying the penalty caused by synchronization delays between source and UE updates. Nevertheless, AoII's reliance on temporal mismatch oversimplifies the evaluation of task completion efficacy. In scenarios like object detection, task-specific metrics (e.g., precision) may better reflect performance than traditional signal-level indicators (e.g., PSNR or BER). Therefore, a comprehensive metric is needed to guide semantic coding and resource allocation in the image semantic transmission system, ensuring efficient onboard processing and transmission [27], [28].

2) Image Semantic Coding: Defining and measuring the information value is a fundamental theoretical challenge in semantic coding [29], [30], which enables the image semantic transmission system to achieve high CR for the massive volume of remote sensing images. Considering the consensus that semantic information represents a latent feature of observable information, the mutual information can be utilized to quantify task-oriented semantics [31], [32], but calculating mutual information for high-dimensional variables remains challenging. Therefore, by utilizing neural networks (NNs) to fit the joint distribution probability function between semantic information and observable information, many deep learning (DL)-based semantic coding schemes are proposed [33], [34]. Initially, these schemes akin to traditional image coding schemes [35], [36], primarily focus on minimizing reconstruction distortion without accounting for the content or task-oriented features of images. With the increasing demand for efficient support of machine vision tasks, coding schemes specifically designed for machine vision have become particularly important. These schemes not only need to ensure compression efficiency but also retain semantic information necessary for task execution [37]. For instance, Hu et al. [38] proposed a scalable coding framework that dynamically allocates coding resources between human perception and machine analysis, achieving remarkable bitrate reduction for machine vision tasks. Considering the region of interest (RoI) of images, Cai et al. [39] proposed an end-to-end optimized RoI image semantic coding scheme, which demonstrates superior task performance compared to traditional image coding schemes. Note that these existing schemes with an RoI are constrained to binary masks, and these DL-based schemes typically require training separate models for different code rates, posing challenges for satellite onboard deployment. Furthermore, Song et al. in [40] proposed an image coding scheme with a single model, which covers a wide range of code rates controlled by quality maps. Although this scheme does not consider how to adjust quality maps for specific tasks and their corresponding impact on the freshness and task completion of UEs, it nonetheless provides valuable insights for the design of image semantic coding for satellite-integrated Internet.

3) Resource Allocation Policy: Nonorthogonal multiple access (NOMA) enables simultaneous communication for multiple UEs in the same frequency band, enhancing the reliability and efficiency of semantic transmission with optimized resource allocation in satellite-integrated Internet [41], [42]. Moreover, optimizing resource allocation to enhance the freshness and keep task updates matching between the source and UEs is increasingly challenging in the complex transmission environment of satellite-to-terrestrial link, particularly due to the impact of time-varying channels and limited onboard resources [43]. Furthermore, the optimization of resource allocation presents a long-term, nonconvex problem, and the Lyapunov optimization framework offers an effective approach to addressing such nonconvex problems by converting them into convex forms, thereby enhancing system

stability [44], [45]. Nevertheless, these existing works usually simplify the information onboard processing and ignore the task completion of UEs after receiving the information.

B. Motivations and Contributions

Satellite-integrated Internet faces critical challenges in transmitting massive remote sensing images under constrained satellite-terrestrial link capacity. Conventional compression schemes with low CRs and independent source-channel coding architectures with cliff effect fail to meet the timeliness and accuracy requirements for mission-critical applications. To address these challenges, we propose a task-oriented semantic coding and utility-optimal transmission (TUT) framework. Unlike conventional methods that focus on pixel-level reconstruction or separate source and channel optimization, our TUT framework emphasizes task-relevant semantic features and unifies their extraction, coding, and transmission under a joint design principle. Our TUT framework systematically integrates JSCC to efficiently extract task-oriented semantic information while maintaining robust performance under low SNR conditions, combined with adaptive resource allocation to ensure information freshness and transmission efficiency. This dual-mechanism directly addresses the fundamental conflict between massive image generation and limited transmission capacity in satellite-integrated Internet. Our contributions to this article are summarized as follows.

1) We propose a novel task-aware metric termed utility loss of information (UoI), which simultaneously captures the freshness, task updates, and task completion status of UEs. By leveraging this metric, our unified semantic coding network enables flexible semantic extraction for diverse tasks and adaptive code rate control. Subsequently, we formulate a long-term average UoI minimization problem and decompose it via the Lyapunov optimization framework, which is solved by utilizing a deep reinforcement learning (DRL)-based proximal policy optimization (PPO) algorithm, leading to UoI-optimal NOMA resource allocation for corresponding UEs.
2) To realize UoI-driven semantic coding, we propose a perceptual-weight map-guided image semantic coding (PMSC) network based on spatial feature transform (SFT). The PMSC network covers a wide range of code rates controlled by arbitrary pixel-level PM through a unified network, making it highly suitable for onboard image processing. By generating optimized PMs and dynamically adjusting the numerical distribution of PM for specific tasks, the PMSC network can perform optimal semantic coding to achieve minimum UoI. Even in the scenario when task updates failed or without optimized PMs, by incorporating the semantic loss of task, our PMSC network can still achieve better task-oriented semantic coding with a single-value PM tailored to UoI than the related schemes.
3) Considering resource constraints, we formulate a long-term average UoI minimization problem for NOMA transmission under power and storage constraints. Given the problem's nonconvexity, we construct two virtual queues for power and storage, then decompose the multislot optimization into a sequence of single-slot Lyapunov drift-plus-penalty (DPP) upper-bounded subproblems. These subproblems are cast as a Markov decision process (MDP), enabling the design of a UoI-optimal resource allocation (URA) policy with our PMSC network, termed PMSC-URA. Simulations demonstrate that the PPO-based PMSC-URA scheme outperforms DDPG-based and MAF benchmarks, achieving minimal long-term average UoI and power consumption.

The remainder of this article is organized as follows. Section II describes the system model and workflow of the TUT framework in satellite-integrated Internet and details the definition of UoI. Section III elaborates on the workflow, loss function design, architecture, and training process of the PMSC network. In Section IV, we elaborate on the UoI optimization problem, transform the problem into a Lyapunov DPP minimization upper bound problem, and propose a PMSC-URA scheme with a PPO algorithm to solve it. Section V presents the simulation results along with the corresponding analysis. The conclusion is presented in Section VI. The main notations used throughout the article are summarized in Nomenclature.

II. System Model

This section outlines the system model and workflow of the TUT framework in satellite-integrated Internet, along with a thorough introduction to UoI.

A. TUT Framework

Our TUT framework in satellite-integrated Internet is depicted in Fig. 1, which includes a semantic empowered satellite S and multiple ground UEs. Specifically, the satellite is equipped with image-processing capabilities (e.g., object detection and image semantic coding) and it employs a hybrid access method in its multibeam coverage area to mitigate interference, which uses NOMA in the same beam for multiple UEs and OMA across different beams. Assume that there are K UEs per beam, denoted as $U = \{\mathrm{UE}_1, \mathrm{UE}_2, \ldots, \mathrm{UE}_K\}$. The transmission period is split into T time slots, each lasting τ, which is greater than twice the propagation delay from S to UEs, and channels are considered quasistationary over one time slot. Additional parameters for S include: carrier frequency $f_c = 30$ GHz, altitude $h = 600$ km, beam diameter $d = 100$ km, and minimum elevation angle 30° [48]. The velocity of S is $v_S = 7.5$ km/s. Given that the negligible UE distance relative to the altitude of S, an equal distance from S to each UE is assumed, leading to identical Doppler shifts for UEs within the same beam, and the protection bandwidth of S is set to twice the Doppler shifts to mitigate their impact on the system [49].

In our TUT framework, the source task updates process is modeled as an N-state discrete Markov chain $(D(t))_{t\in\mathbb{N}}$, where $D(t) \in \{1, 2, \ldots, N\}$, as depicted in the lower-right corner of

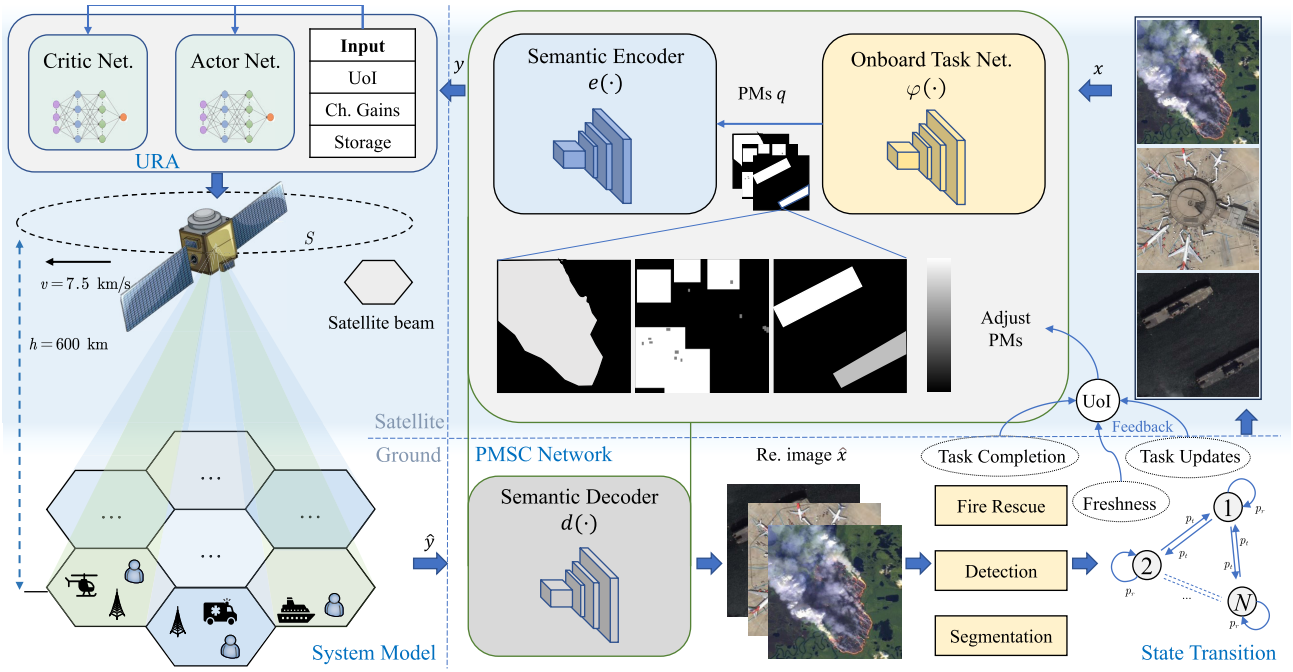


Fig. 1. Our TUT framework in satellite-integrated Internet mainly consists of four components: the system model, the PMSC network, the PPO-based URA policy, and the state transition process. The system model presents the satellite multicasts to multiple UEs via NOMA transmission. The PMSC network is composed of onboard and ground parts. The onboard part first extracts the semantic information $\boldsymbol{y}$ from the image $\boldsymbol{x}$, which is captured by S upon detecting a task update. Next, using the URA policy that accounts for UoI, channel gains, and the storage capacity of S, $\boldsymbol{y}$ is transmitted to the ground UEs. Upon UEs receiving $\hat{\boldsymbol{y}}$, the ground part of the PMSC network reconstructs $\hat{\boldsymbol{x}}$. At the UE side, the reconstruction of $\hat{\boldsymbol{x}}$ and corresponding task completion collectively form a new task update.

Fig. 1. The probabilities of remaining in the current task state and transitioning to another task state are defined as p_r and $p_t = (1 - p_r)/(N - 1)$, respectively, with state transitions occurring at the start of each time slot t. When S detects a task update for UE_i in time slot t, it employs the sample-at-change (SAC) strategy, sampling and transmitting semantic information of the corresponding image $\boldsymbol{x}_i(t)$ for UE_i. Then, the onboard task network $\varphi(\cdot)$ processes the image $\boldsymbol{x}_i(t)$ and generates the corresponding task-oriented PM $\boldsymbol{q}_i(t)$ as follows:

$$\boldsymbol{q}_i(t) = \varphi(\boldsymbol{x}_i(t)) \tag{1}$$

which is controlled and adjusted by UoI in the following.

Subsequently, $\boldsymbol{q}_i(t)$ along with $\boldsymbol{x}_i(t)$ is both used as inputs of the semantic encoder $e(\cdot)$ and then performs semantic coding on $\boldsymbol{x}_i(t)$ based on $\boldsymbol{q}_i(t)$, ultimately extracting the semantic information $\boldsymbol{y}_i(t)$

$$\boldsymbol{y}_i(t) = e\big(\boldsymbol{x}_i(t), \boldsymbol{q}_i(t)\big). \tag{2}$$

Notably, the proposed PMSC network employs a JSCC architecture, where the extracted semantic information is quantized with a learnable entropy model [50], [51] and mapped into channel input symbols $\boldsymbol{y}_i(t)$ for UE_i. The shadowed-Rician fading model, integrating a Nakagami-distributed line-of-sight component and Rayleigh scattering, provides closed-form statistical characterizations for land mobile satellite channels. This hybrid structure effectively captures both shadowing and multipath fading effects [1], [52], while maintaining analytical tractability for system performance evaluation. For the considered network with independent and identically distributed (i.i.d.) fading across UE, the channel gain $|h_i|^2$ follows the probability density function [53]:

$$f_{|h_i|^2}(x) = \left(\frac{2b_i m_i}{2b_i m_i + \Omega_i}\right)^{m_i} \frac{1}{2b_i} \exp\left(-\frac{x}{2b_i}\right) \cdot {}_1F_1\left(m_i, 1, \frac{\Omega_i x}{2b_i(2b_i m_i + \Omega_i)}\right) \tag{3}$$

where $2b_i$, m_i, and Ω_i denote the average power of the scatter component, the Nakagami-m parameter, and the LoS component, respectively. ${}_1F_1(\cdot,\cdot,\cdot)$ represents the Gauss hypergeometric function.

Assume that S is equipped with N_s transmitting antennas, while each UE has a single receiving antenna, and the cumulative distribution function (cdf) of $|\mathbf{h}_i|^2$ for S is described as follows [54]:

$$F_{|\mathbf{h}_i|^2}(x) = 1 - \sum_{j_1=0}^{m_i-1} \cdots \sum_{j_{N_s}=0}^{m_i-1} \sum_{p=0}^{\Lambda_1-1} \frac{\Xi_1(N_s)(\Lambda_1-1)!}{p!} \times (\beta_i - \delta_i)^{-(\Lambda_1-p)} x^p \exp[-(\beta_i - \delta_i)x] \tag{4}$$

where $\Lambda_1 = \sum_{i=1}^{N_s} j_i + N_s$, $\Xi_1(N_s) = (\eta_i)^{N_s} \prod_{j=1}^{N_s} \xi(j_i) \times \prod_{q=1}^{N_s-1} B_f(\sum_{l=1}^{q} j_l + q, j_{q+1} + 1)$, $\xi(j_i) = ((-1)^{j_i}(1 - m_i)_{j_i}(\delta_i x)^{j_i})/(j_i!)^2$, $\eta_i = (2b_i m_i/(2b_i m_i + \Omega_i))^{m_i}/2b_i$, $B_f(\cdot,\cdot)$ denotes the Beta function, $\delta_i = \Omega_i/2b_i/(2b_i m_i + \Omega_i)$, and $\beta_i = 1/2b_i$.

Considering the limited resources of S, the URA policy allocates power resources to UEs in a rational manner, taking into account UoI, channel gains, and storage. Let $\mathbf{p}_i(t) \in \mathbb{C}^K$ denote the corresponding transmit power vector allocated by the URA policy. Accordingly, the superimposed signal

transmitted by S for K UEs is expressed as

$$\mathbf{s}(t) = \sum_{i=1}^{K} \mathbf{p}_i(t)\boldsymbol{y}_i(t). \tag{5}$$

After transmission through the shadow-Rician fading channel, the received signal $\boldsymbol{r}_i(t)$ at UE_i is expressed as follows:

$$\boldsymbol{r}_i(t) = \mathbf{g}_i(t)\mathbf{s}(t) + \boldsymbol{n}_i(t) \tag{6}$$

where $\mathbf{g}_i(t) = \mathbf{h}_i(t)(G_S G_{\mathrm{UE}} L_F^{-1})^{1/2}$ is the joint channel gain. G_S and G_{UE} are the satellite beam gain and antenna gain, respectively. $L_F(\mathrm{dB}) = 92.4{+}20\log f(\mathrm{GHz}){+}20\log d(\mathrm{km})$ is the free space loss from S to UE_i. $\boldsymbol{n}_i(t) \sim \mathcal{CN}(0, \sigma^2\mathbf{I})$ denotes i.i.d. additive white Gaussian noise samples with zero mean and variance σ^2. The channel coefficients remain stationary within a time slot.

Then, UE_i uses successive interference cancellation (SIC) to separate $\boldsymbol{y}_i(t)$ from the received superimposed signal $\boldsymbol{r}_i(t)$. Without loss of generality, we assume the signal power decreases with the UE's index. The decoding process starts with $\boldsymbol{y}_K(t)$, performing maximum likelihood detection on the composite signal under interference from weaker UEs. If separated correctly, $\boldsymbol{y}_K(t)$ is subtracted from $\boldsymbol{r}_i(t)$, and the process continues with $\boldsymbol{y}_{K-1}(t)$ until $\boldsymbol{y}_i(t)$ is recovered at UE_i. The signal-to-interference-and-noise-ratio (SINR) of UE_i is given by [55]

$$\rho_i(t) = \frac{\mathbf{p}_i(t)|\mathbf{g}_i(t)|^2}{\sum_{j=1}^{i-1}\mathbf{p}_j(t)|\mathbf{g}_i(t)|^2 + \kappa\sum_{j=i+1}^{K}\mathbf{p}_j(t)|\mathbf{g}_i(t)|^2 + \sigma^2} \tag{7}$$

where $(\sum_{j=1}^{i-1}\mathbf{p}_j(t)|\mathbf{g}_i(t)|^2)$ is the interference caused by undecoded UEs. $(\kappa\sum_{j=i+1}^{K}\mathbf{p}_j(t)|\mathbf{g}_i(t)|^2)$ is the residual component of decoded UEs and $\kappa \in [0, 1]$ is the error-propagation factor. Let B denote channel bandwidth, the data rate of UE_i is then represented as follows:

$$R_i(t) = B\log_2(1 + \rho_i(t)). \tag{8}$$

The extracted semantic information $\boldsymbol{y}_i(t)$ is transmitted from S to UE_i over one or more time slots, depending on the volume of $\boldsymbol{y}_i(t)$ and $R_i(t)$. Upon the transmission of $\boldsymbol{y}_i(t)$ accomplishing, the separated signal $\hat{\boldsymbol{y}}_i(t)$ by SIC is fed into the semantic decoder $d(\cdot)$ of the PMSC network, which reconstructs the image $\hat{\boldsymbol{x}}_i(t)$

$$\hat{\boldsymbol{x}}_i(t) = d\big(\hat{\boldsymbol{y}}_i(t)\big). \tag{9}$$

After image reconstruction, UE_i performs the corresponding tasks, such as fire rescue, object detection, image semantic segmentation, and so on. At the UE side, the reconstruction of $\hat{\boldsymbol{x}}_i(t)$ and corresponding task completion constitute a task update for UE_i. Let $D_i(t)$ and $\hat{D}_i(t)$ represent the current state of source and perceived task state of UE_i, respectively. A mismatch, indicated by $D_i(t) \neq \hat{D}_i(t)$, occurs if the UE fails to receive the semantic information $\boldsymbol{y}_i(t)$ in time or complete the task. Upon successfully updating the state, the UE sends an acknowledgment (ACK) to S; otherwise, it sends a negative ACK (NACK). In the case of NACK, S will retransmit the unrecovered semantic information $\boldsymbol{y}_i(t)$ in the next time slot unless the task state for UE_i changes before retransmission.

B. Definition of UoI

Existing metrics, such as AoI, predominantly concentrate on temporal freshness, thereby neglecting the semantic aspects of task-oriented communication systems. In contrast, the UoI metric effectively bridges this gap by integrating traditional communication metrics with task-oriented requirements and provides a unified framework for optimizing semantic-aware systems. Specifically, UoI simultaneously captures the freshness of information, task updates, and task completion status of UEs, which can be expressed as a specific mapping using composite function [45]

$$\big(f(t), g\big(D(t), \hat{D}(t)\big), C(D(t))\big) \in \mathcal{T}\times\mathcal{X}\times\mathcal{C} \xrightarrow{\mathcal{F}} \Delta(t). \tag{10}$$

The first attribute, $f(t) \in \mathcal{T}$, akin to the AoI metric, represents the freshness function, which typically is defined as the time elapsed since the last successful update was received by the UE. For instance, $f_i(t) = t - t_m$, where t_m denotes the most recent time slot before t when the UE successfully received an update from the source. $g(D(t), \hat{D}(t)) \in \mathcal{X}$ represents the task state error detection function, which can be defined using measures such as the mean squared error (mse) or an indicator function. Meanwhile, $C(t) \in \mathcal{C}$ represents the task completion status function. While the core structure of UoI remains consistent, the definition of $C(t)$ varies across tasks to reflect their unique objectives. For instance, in a classification task, the task completion term $C(t)$ can be defined using classification accuracy or the $F1$-score, then the UoI can be expressed as follows:

$$\Delta_i(t) = \sum_{\epsilon=t_m+1}^{t} (1 - \mathrm{Accuracy}(\epsilon)) \cdot \mathbb{1}_{D_i(t)\neq\hat{D}_i(t)} \tag{11}$$

where $\mathrm{Accuracy}(t)$ quantifies the classification correctness at time t and $\mathbb{1}_{D_i(t)\neq\hat{D}_i(t)}$ denotes the indicator function. For an object detection task, this function can be set to reflect the precision of the UE's object detection after receiving the image $\hat{\boldsymbol{x}}$. In this context, the UoI for UE_i can be defined as follows:

$$\Delta_i(t) = \sum_{\epsilon=t_m+1}^{t} (1 - \mathrm{mAP}(\epsilon)) \cdot \mathbb{1}_{D_i(t)\neq\hat{D}_i(t)} \tag{12}$$

where $\mathrm{mAP}(\epsilon)$ represents the mean average precision (mAP) metric for object detection evaluated at time slot ϵ, with its formal definition being omitted here for space considerations. The complete computational procedure can be found in [56].

The information transmission requirements in each time slot and the resources allocation policy of TUT both influence the UoI. The long-term average UoI, $\bar{\Delta}_{\mathrm{UoI}}$, is defined to evaluate the overall performance of the TUT framework and is expressed as

$$\bar{\Delta}_{\mathrm{UoI}} = \lim_{T\to\infty}\frac{1}{T}\cdot\frac{1}{K}\mathbb{E}\left[\sum_{i=1}^{K}\sum_{t=1}^{T}\Delta_i(t)\right]. \tag{13}$$

III. Image Semantic Coding

In this section, we introduce the image coding process, loss function design, architecture, and pipeline of the PMSC network.

A. Image Coding Process of the PMSC Network

As illustrated in the central portion of Fig. 1, the proposed PMSC network comprises three primary modules: the semantic encoder $e(\cdot)$, the semantic decoder $d(\cdot)$, and the onboard task network $\varphi(\cdot)$. S acquired images are treated as observable information X, which maintains a probabilistic relationship with the semantic information Y. The purpose of $\varphi(\cdot)$ is to generate the PM $\boldsymbol{q}$, guiding $e(\cdot)$ in performing semantic coding of the image. Specifically, the method for extracting $\boldsymbol{q}$ using $\varphi(\cdot)$ depends on the specific task or generation strategy. For instance, if the goal is to assign different PM values to distinct target objects within an image, $\varphi(\cdot)$ can be designed as an object detection network.

When a new task update is generated at the source during time slot t for UE_i, the transmitter samples an image $\boldsymbol{x}_i(t)$ from the observation space X. The onboard task network $\varphi(\cdot)$ then outputs the PM $\boldsymbol{q}_i(t)$ corresponding to $\boldsymbol{x}_i(t)$. The semantic encoder $e(\cdot)$ performs semantic coding of $\boldsymbol{x}_i(t)$ with adjustable code rates in different regions based on $\boldsymbol{q}_i(t)$, which is controlled and adjusted by UoI. The encoded semantic information $\boldsymbol{y}_i(t) = e(\boldsymbol{x}_i(t), \boldsymbol{q}_i(t))$ is transmitted through the channel and received by UE_i. Upon receiving $\boldsymbol{r}_i(t)$, the semantic decoder $d(\cdot)$ decodes the semantic information to reconstruct the image, and the reconstructed image $\hat{\boldsymbol{x}}$ is used for downstream tasks such as object detection, image segmentation, and so on.

In our PMSC network, both $e(\cdot)$ and $d(\cdot)$ are implemented using NNs. During training, $e(\cdot)$ and $d(\cdot)$ are jointly trained using the library dataset $\mathcal{D} = (X, Z)$, which consists of the set of observable information X and the corresponding task Z. During inference, the two modules are pretrained and fixed for a specific downstream task. It is important to note that $e(\cdot)$ and $d(\cdot)$ are entirely independent of the onboard task network $\varphi(\cdot)$. This means that for different tasks, such as object detection and image semantic segmentation, the semantic encoder and the decoder do not need to be retrained, which significantly enhances the adaptability and practicality of the PMSC network for deployment on satellites.

B. Loss Function Design of the PMSC Network

Traditional image transmission systems typically evaluate image reconstruction performance using metrics like the PSNR and MS-SSIM, with the coding process optimized based on rate-distortion theory. However, the primary goal of image semantic transmission is to convey the task-oriented features of images rather than to transmit individual pixels with precision. In other words, a fundamental difference in image semantic transmission lies in its requirement for not only image recovery performance but also additional semantic performance tailored to specific tasks. This means that the image semantic transmission system should prioritize the importance of transmitted information, ensuring key information is recovered with high semantic fidelity and perceptual quality, while less critical information may be discarded or transmitted at lower quality.

Existing lossy image coding schemes leveraging DL typically adopt a transform coding approach, which maps pixel data to a quantized latent representation, followed by lossless coding of the representation. These schemes are designed to minimize both the expected bit-stream length and the expected distortion of the reconstructed image relative to the original, leading to a rate-distortion optimization problem

$$\min R_c + \lambda \cdot D_c \tag{14}$$

where R_c denotes the code rate constraint of the encoder, D_c represents the distortion constraint, and λ is the Lagrange multiplier governing the rate-distortion tradeoff. The rate term R_c could be materialized as code rate R or other rate-related metrics, and the distortion term D_c may correspond to a closed-form likelihood metric, such as mse or SSIM.

In the existing task-oriented image semantic transmission systems, semantic distortion can be evaluated through practical tasks, such as object detection, semantic segmentation, pose estimation, or action recognition. Notably, although these tasks objectively evaluate semantic distortion, the definition of semantic distortion varies across tasks.

For object detection tasks, let $\phi_{\mathrm{obj}}(\cdot)$ represent the object detection network. The prediction results for the input image $\boldsymbol{x}$ and the reconstructed image $\hat{\boldsymbol{x}}$ are denoted as $\phi_{\mathrm{obj}}(\boldsymbol{x})$ and $\phi_{\mathrm{obj}}(\hat{\boldsymbol{x}})$, respectively. Here, the semantic loss L_{sem} is defined as the accuracy of $\phi_{\mathrm{obj}}(\cdot)$ on the reconstructed image $\hat{\boldsymbol{x}}$, using the ground-truth (GT) semantic labels of $\boldsymbol{x}$. The intersection over union (IoU) is employed to measure the accuracy of object detection during training and is defined as

$$\begin{aligned}&\mathrm{IoU}\left(\phi_{\mathrm{obj}}(\boldsymbol{x})\right), \phi_{\mathrm{obj}}\left(\hat{\boldsymbol{x}}\right)\big)\big)\\ &= \mathrm{overlap}\left(\boldsymbol{B}_o^{n_c}, \boldsymbol{B}_r^{n_c}\right)/\mathrm{union}\left(\boldsymbol{B}_o^{n_c}, \boldsymbol{B}_r^{n_c}\right)\end{aligned} \tag{15}$$

where $\phi_{\mathrm{obj}}(\boldsymbol{x}) = B_o^{n_c}$ represents the GT bounding box set of class n_c for $\boldsymbol{x}$ and $\phi_{\mathrm{obj}}(\hat{\boldsymbol{x}}) = B_r^{n_c}$ represents the predicted bounding box set of class n_c for $\hat{\boldsymbol{x}}$. Note that IoU values range from 0 to 1. Then, we adopt IoU loss as the semantic loss for object detection which is expressed as

$$L_{\mathrm{sem}} = 1 - \frac{1}{N_c}\sum_{n=1}^{N_c} \mathrm{IoU}\left(\boldsymbol{B}_g^{n_c}, \boldsymbol{B}_d^{n_c}\right) \tag{16}$$

where N_c denotes the total number of classes in $\boldsymbol{x}$. Regarding the IoU-based loss function discussed here, its primary purpose is to illustrate the concept of integrating semantic loss into the model training framework, the design of semantic loss functions remains an open research area that warrants further investigation.

The loss functions of most existing image semantic coding systems based on NNs, typically include both semantic loss and pixel loss [57]. Our proposed loss function also follows this dual-structure. In other words, compared to the traditional rate-distortion optimization problem, the semantic rate-distortion problem introduces an additional semantic constraint, which is expressed as

$$L = R_c + \lambda \cdot (D_c + \mu \cdot L_{\mathrm{sem}}) \tag{17}$$

where μ is the weighting parameter balancing the observable and semantic information.

Additionally, our PMSC network achieves variable-rate semantic coding by minimizing the objective function

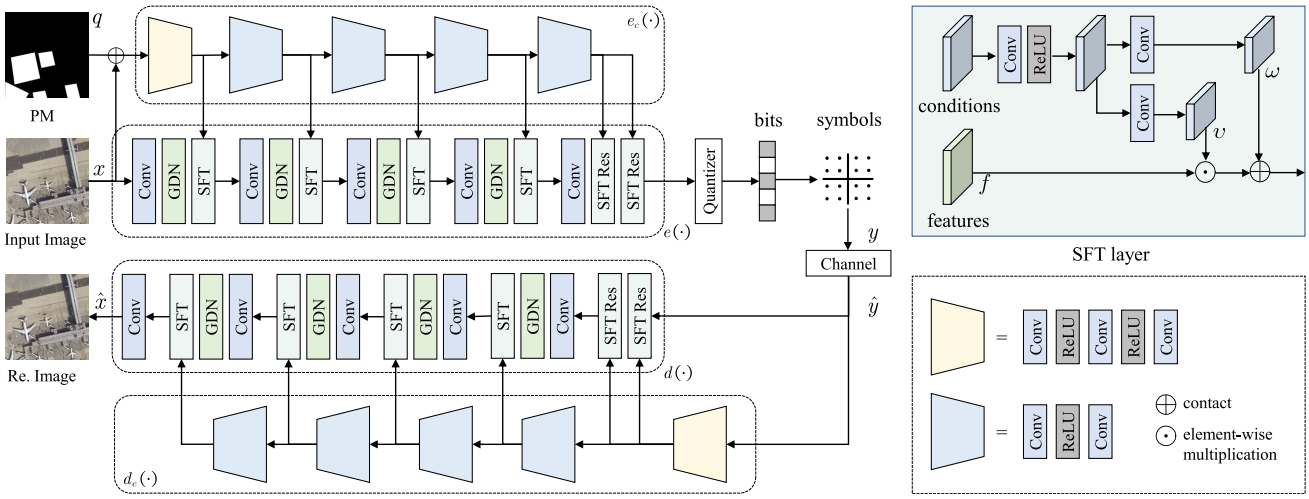


Fig. 2. Architecture of the PMSC network and the SFT layer. We employ the SFT layers in the semantic encoder and utilize several condition networks to generate prior condition features from the input image and the PM. The GDN denotes the generalized divisive normalization.

$R_c + \Lambda^T(D + \mu \cdot L_{\text{sem}})$, where $\Lambda = [\lambda_i]_{i=1:N_\Lambda} \in \mathbb{R}^{N_\Lambda}$ is a vector of Lagrange multipliers that vary with the priority map (PM) $\boldsymbol{q}$.

Specifically, in the PMSC network, the code rate R is implicitly controlled through the Lagrange multipliers $\Lambda = [\lambda_i]$, which are directly linked to the PM $\boldsymbol{q}$. Each multiplier $\lambda_i \in \Lambda$ is determined by its corresponding element $q_i \in \boldsymbol{q}$ via a predefined monotonic function $F : [0, 1] \to \mathbb{R}^+$. This design ensures that regions with higher q_i values (indicating greater semantic importance) are assigned larger λ_i values. Consequently, stronger regularization is applied to R_c for critical regions, enabling adaptive bit allocation to preserve semantic fidelity. By leveraging the monotonic relationship between q_i and λ_i, the model dynamically allocates more bits to semantically significant regions, optimizing the tradeoff between coding efficiency and semantic preservation.

C. Architecture and Pipeline of the PMSC Network

The proposed PMSC network incorporates an SFT module to learn the mapping function $F_{\text{map}}(\upsilon, \omega)$, which generates modulation parameters based on the prior condition Ψ derived from the PM $\boldsymbol{q}$. Fig. 2 depicts the detailed architecture of the SFT module. The SFT module is trained to generate a set of element-wise affine parameters (υ, ω) for the intermediate feature f, which are influenced by the external condition Ψ. Thus, the feature transformation in the SFT layer is expressed as

$$\text{SFT}(f, \Psi) = \upsilon \odot f + \omega. \tag{18}$$

First, the image $\boldsymbol{x}$ is encoded using the semantic encoder $e(\cdot)$ and the condition network $e_c(\cdot)$ based on the PM $\boldsymbol{q}$, resulting in the latent representation $\boldsymbol{y}$

$$\boldsymbol{y} = e(\boldsymbol{x}, \Psi_1), \text{ where } \Psi_1 = e_c(\boldsymbol{x}, \boldsymbol{q}). \tag{19}$$

The latent representation $\boldsymbol{y}$ encapsulates spatially adaptive quality information. Next, $\boldsymbol{y}$ is transmitted through a shadow-Rician channel and received by the UEs. To reconstruct the image, the semantic decoder $d(\cdot)$ and the condition network $d_c(\cdot)$ process the received $\hat{\boldsymbol{y}}$, as described by

$$\hat{\boldsymbol{x}} = d(\boldsymbol{x}, \Psi_2), \text{ where } \Psi_2 = d_c(\hat{\boldsymbol{y}}). \tag{20}$$

Algorithm 1 Training the PMSC Network

Input: Library dataset $\mathcal{D} = (X, Z)$, pretrained onboard task network $\phi(\cdot)$, pretrained downstream task network of UEs $\varphi(\cdot)$, training epoch E_p, learning rate lr, weight parameters λ and μ, batch size b.

Output: Neural network parameters for $e(\cdot)$, $d(\cdot)$.

1: **for** epoch $= 1 : E_p$ **do**
2: Sample a batch of images $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_b \in X$;
3: Obtain b PMs $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_b$ for the images $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_b$ using $\phi(\cdot)$;
4: Semantic coding images $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_b$ with PMs $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_b$ using $e(\cdot)$;
5: Transmit the coded data to the UEs through the shadow-Rician channel;
6: Reconstruct the images $\hat{\boldsymbol{x}}_1, \ldots, \hat{\boldsymbol{x}}_b$ using $d(\cdot)$;
7: Operate task using the reconstructed images $\hat{\boldsymbol{x}}_1, \ldots, \hat{\boldsymbol{x}}_b$ by $\phi(\cdot)$;
8: Calculate the gradients $\nabla_{\theta_e, \theta_d} L$ and update $e(\cdot), d(\cdot)$.
9: **end for**
10: **return** The PMSC network parameters $e(\cdot), d(\cdot)$.

D. Training Algorithm

The value of semantic information in images varies depending on the content and its task-oriented features. In the training process, only the semantic information relevant to the task requires high-precision transmission. Therefore, the goal of training semantic encoders and decoders is to ensure accurate coding without sacrificing semantic integrity, while minimizing redundancy in the image.

While training, the reconstructed image $\hat{\boldsymbol{x}}$ is input into the downstream task network to obtain the corresponding task output $\hat{\boldsymbol{z}}$. The original image $\boldsymbol{x}$ and its corresponding output $\boldsymbol{z}$ are known for the receiver during the training process.

Thus, a complete training sample is defined as the tuple $(\boldsymbol{x}, \hat{\boldsymbol{x}}, z, \hat{z})$, and the collection of such samples constitutes a complete training batch. The training process is summarized in Algorithm 1.

Since the quantization operation is inherently nondifferentiable when mapping and quantizing the semantic features into channel input symbols $\boldsymbol{y}$, this poses a challenge for back-propagation during training. To address this issue, we replace the actual quantization operation with the addition of uniform noise during the training phase [58]. This approach provides a differentiable approximation of the quantization process, enabling gradient-based optimization. During the evaluation phase, we revert to using the actual quantization operation to ensure consistency with deployment conditions.

IV. UoI Minimization Problem

This section starts with the formulation of the long-term average UoI minimization problem, considering two long-term and one short-term constraint in the TUT framework. Subsequently, we transform the UoI optimization problem into an upper-bound minimization problem of the Lyapunov DPP term by constructing two virtual queues, utilizing Lyapunov optimization theory, and solving it with our PMSC-URA scheme.

A. Problem Formulation

The optimization of UoI depends on three key factors: information freshness, task state updates, and task completion status. Under normal conditions, lower PM values enable higher CR, reducing transmission overhead and improving information freshness and state updates. While aggressive compression enhances transmission efficiency, it may degrade image quality at the receiver, negatively impacting task completion accuracy. To achieve optimal UoI, the TUT framework employs adaptive PM distribution adjustment based on real-time feedback. When UoI degradation is detected, the system identifies the dominant factor (e.g., failed updates versus failed tasks) and optimizes PM for either higher compression or bit allocation, achieving multiobjective balance to minimize UoI across all dimensions. Furthermore, optimizing UoI for the TUT framework in satellite-integrated Internet also requires consideration of the following constraints.

1) Short-Term Peak Power Constraint: Given the limited power resources of satellites, power allocation must account for short-term peak power constraints. Let $P_{\max}$ represent the maximum total power that satellite S can allocate to all UEs in time slot t, as follows:

$$\sum_{i=1}^{K} |\mathbf{p}_i(t)|^2 \leq P_{\max}. \tag{21}$$

2) Long-Term Average Power Constraint: Let P_{mean} represent the long-term average power that satellite S can allocate to all UEs. The long-term average power constraint can be expressed as follows:

$$\lim_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\left[\sum_{i=1}^{K} |\mathbf{p}_i(t)|^2\right] \leq P_{\text{mean}}. \tag{22}$$

3) Storage Constraint: Additionally, the storage resources of satellite S are constrained. We denote $Q_i(t)$ as the data packets waiting to be transmitted to UE_i at time slot t, stored temporarily in the queue backlog $\mathcal{Q}_i$. At the start of each time slot t, if task updates and a new image is sampled for UE_i, the encoded semantic information $\boldsymbol{y}_i(t)$ arrives at $\mathcal{Q}_i$ according to $A_i(t)$, where $A_i(t)$ is the arrive data rate determined by the volume of $\boldsymbol{y}_i(t)$. Once UE_i successfully recovers the image $\hat{\boldsymbol{x}}_i(t)$ through SIC, it leaves $\mathcal{Q}_i$. Therefore, to prevent data overflow, the following storage constraint must be satisfied:

$$\lim_{T\to\infty} \frac{1}{T} \mathbb{E}[|Q_i(T)|] = 0. \tag{23}$$

Hence, the specific long-term average UoI minimization problem (P1) can be formulated as

$$\text{P1: } \min \ \bar{\Delta}_{\text{UoI}} \tag{24a}$$

$$\text{s.t. } C1: \sum_{i=1}^{K} |\mathbf{p}_i(t)|^2 \leq P_{\max} \tag{24b}$$

$$C2: \lim_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\left[\sum_{i=1}^{K} |\mathbf{p}_i(t)|^2\right] \leq P_{\text{mean}} \tag{24c}$$

$$C3: \lim_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[|Q_i(t)|] = 0, i \in \{1, 2, \ldots, K\}. \tag{24d}$$

B. Lyapunov Optimization

Due to its long-term nature across multiple time slots, the optimization problem in (24) presents significant challenges for direct solutions. We apply the DPP algorithm in the Lyapunov optimization framework to transform the long-term optimization problem, subject to the constraints $C2$ and $C3$, into a solvable single-slot problem by introducing power debt and data buffer queues. The evolutions of the two virtual queues are described as follows.

1) Power Debt Queue: A power debt queue $P(t)$ is employed to regulate the average power usage in each time slot, ensuring it stays under P_{mean}, and we have

$$P(t+1) = \max\left(P(t) + \sum_{i=1}^{K} |\mathbf{p}_i(t)|^2 - P_{\text{mean}}, 0\right). \tag{25}$$

2) Data Buffer Queue: The data buffer queue $Q_i(t)$ for UE_i in S is monitored. This queue represents the accumulated backlog during time slot t, and its evolution is characterized as follows:

$$Q_i(t+1) = \max(Q_i(t) + A_i(t) - \tau R_i(t), 0). \tag{26}$$

For long-term optimization problems, operations that satisfy long-term constraints need to be decomposed into single-slot decisions. The core idea of applying Lyapunov optimization theory is to directly optimize the delay objective in each time slot while maintaining the constraints necessary for queue stability. Let $V(t)$ represent the virtual queue state at time slot t

$$V(t) = [P(t), [Q_1(t), \ldots, Q_K(t)]]. \tag{27}$$

Accordingly, the quadratic Lyapunov function is formulated as

$$L(V(t)) \triangleq \frac{1}{2}\left[P^2(t) + \sum_{i=1}^{K} Q_i^2(t)\right]. \tag{28}$$

The Lyapunov drift characterizes the variation in (28) between time slots t and $t+1$, defined as the expected value conditioned on the current state $V(t)$

$$\mathrm{LD}(V(t)) = \mathbb{E}[L(V(t+1)) - L(V(t)) \mid V(t)]. \tag{29}$$

By minimizing $\mathrm{LD}(V(t))$ in a limited range, we can effectively control congestion and ensure the stability of the two virtual queues (25) and (26). Taking measures to minimize its bounds can stabilize the system, but this may result in unnecessarily large UoI. To minimize the long-term average UoI, instead of solely minimizing the conditional Lyapunov drift bounds, we minimize the Lyapunov DPP bounds. The Lyapunov penalty term related to UoI is defined as follows:

$$\mathrm{LP}(V(t)) = \frac{1}{K}\mathbb{E}\left[\sum_{i=1}^{K} \Delta_i\left(D_i(t), \hat{D}_i(t), t\right) \mid V(t)\right]. \tag{30}$$

Therefore, the optimization objective is the DPP term of the TUT framework, which is given by

$$\mathrm{DPP}(V(t)) \triangleq \mathrm{LD}(V(t)) + v_p \cdot \mathrm{LP}(V(t)) \tag{31}$$

where v_p is the weight of the penalty term, with $v_p \geq 0$. Based on the Lyapunov optimization framework, the upper bound of the DPP term can be derived as follows:

$$\begin{aligned}
&\mathrm{DPP}(V(t)) \\
&\leq\ c + v_p \cdot \frac{1}{K}\mathbb{E}\left[\sum_{i=1}^{K} \Delta_i\left(D_i(t), \hat{D}_i(t), t, t_m\right) \mid V(t)\right] \\
&\quad + \mathbb{E}\left[P(t) \cdot \left(\sum_{i=1}^{k} |\mathbf{p}_i(t)|^2 - P_{\mathrm{mean}}\right) \mid V(t)\right] \\
&\quad + \mathbb{E}\left[\sum_{i=1}^{K} Q_i(t) \cdot (A_i(t) - R_i(t)) \mid V(t)\right]
\end{aligned} \tag{32}$$

where c is a constant.

The long-term problem (P1) is now reformulated as a single-slot optimization problem (P2) by focusing on minimizing the upper bound of the DPP term

$$\begin{aligned}
\text{P2: } \min\ & P(t) \cdot \sum_{i=1}^{K} |\mathbf{p}_i(t)|^2 - \sum_{i=1}^{K} Q_i(t) \cdot R_i(t) \\
& + v_p \cdot \frac{1}{K}\mathbb{E}\left[\sum_{i=1}^{K} \Delta_i\left(D_i(t), \hat{D}_i(t), t, t_{\mathrm{m}}\right) | V(t)\right] \\
\text{s.t.} & \sum_{i=1}^{K} |\mathbf{p}_i(t)|^2 \leq P_{\max}.
\end{aligned} \tag{33}$$

C. PMSC-URA Scheme

Traditional optimization algorithms, designed for static environments, are inadequate for addressing the complexities of satellite-to-terrestrial links due to the dynamic nature of the environment. The PPO algorithm, based on DRL, can effectively address this issue by directly interacting with the environment to derive optimal strategies in dynamic settings. Consequently, we present a solution to obtain the URA policy by employing the PPO algorithm in conjunction with PMSC. First, we convert the optimization problem (P2) into an MDP. MDP can be characterized by the quintuple $\{\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma\}$, where $\mathcal{S}$, $\mathcal{A}$, $\mathcal{P}$, $\mathcal{R}$, and γ represent the state space, action space, state–action transition probability matrix, reward function, and discount factor, respectively. The detailed elements of the established quintuple MDP are presented as follows.

1) State: The state of MDP at time slot t is defined as $s_t = \{s_t^1, s_t^2, s_t^3\}$, $s_t \in \mathcal{S}$, where $s_t^1 = \{\Delta_1, \Delta_2, \ldots, \Delta_K\}$, $s_t^2 = \{Q_1, Q_2, \ldots, Q_K\}$, and $s_t^3 = \{\mathbf{g}_1, \mathbf{g}_2, \ldots, \mathbf{g}_K\}$. The state simultaneously integrates UEs' UoI, buffer queues, and channel states, which collectively enhance URA's adaptability to highly dynamic environments.

2) Action: The action of MDP at time slot t is defined as power allocation for all UEs $a_t = \{|\mathbf{p}_1|^2, |\mathbf{p}_2|^2, \ldots, |\mathbf{p}_K|^2\}$, and $a_t \in \mathcal{A}$.

3) Transition Probability: $\Pr(s_{t+1}|s_t, a_t)$ denotes the probability of transforming state from s_t to s_{t+1} when taking action a_t, and all $\Pr(s_{t+1}|s_t, a_t)$ make up $\mathcal{P}$.

4) Reward: The reward $r_{t+1}(s_t, a_t)$ in our TUT framework is formulated as the change in DPP value resulting from executing action a_t in state s_t. Furthermore, if the short-term peak power constraint $P_{\max}$ is violated, the reward is assigned as the negated Lyapunov penalty term $(-\mathrm{LP}(V(t)) \leq 0)$." Thus, $r_{t+1}(s_t, a_t)$ is expressed as follows:

$$\begin{aligned}
&r_{t+1}(s_t, a_t) \\
&= \begin{cases} \mathrm{DPP}(V(t-1)) - \mathrm{DPP}(V(t)), & \sum_{i=1}^{K} |\mathbf{p}_i|^2 \leq P_{\max} \\ -\mathrm{LP}(V(t)), & \text{otherwise.} \end{cases}
\end{aligned} \tag{34}$$

5) Discount Factor: $\gamma \in [0, 1]$ balances the long- and short-term benefits of the TUT framework in MDP. The long-term reward with a discount can be expressed as

$$\mathbf{R}_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \cdots = \sum_{\tau=0}^{\infty} \gamma^k r_{t+\tau+1}. \tag{35}$$

The PPO algorithm adopts an actor–critic architecture, comprising both actor and critic networks. Let θ_a and θ_c represent the parameters of the actor and critic networks, respectively.

Actor: The actor selects action a_t at time slot t given the state s_t according to the strategy $\pi_{\theta_a}(a_t|s_t)$. Since the state-space $\mathcal{S}$ is continuous, the Gaussian distribution is used to parameterize the strategy

$$\pi_{\theta_a}(a_t|s_t) := \mathcal{N}\left(\mu_{\theta_a}, \sigma_{\theta_a}^2\right) \tag{36}$$

where μ_{θ_a} and $\sigma_{\theta_a}^2$ represent the mean and variance of Gaussian distribution, respectively.

The PPO algorithm is prone to greedy behavior, and large-scale policy updates may result in performance degradation. To address this, a surrogate objective function is introduced during the parameters update process to maintain

consistent policy improvement. The loss function of the actor network is defined as

$$L(\theta_a) = \hat{\mathbb{E}}_{\pi_{\theta_a'}}\left[\frac{\pi_{\theta_a}(a_t|s_t)}{\pi_{\theta_a'}(a_t|s_t)} \cdot \hat{\mathbf{A}}_t^{\pi_{\theta_a'}}\right] \tag{37}$$

where $\pi_{\theta_a'}(a_t|s_t)$ represents the old policy, and $\hat{\mathbf{A}}_t^{\pi_{\theta_a'}}$ is an estimate of the advantage function, which can be expressed as

$$\hat{\mathbf{A}}_t^{\pi_{\theta_a'}}(s_t, a_t) = \hat{\mathbf{Q}}_t^{\pi_{\theta_a'}}(s_t, a_t) - \hat{\mathbf{V}}_t^{\pi_{\theta_a'}}(s_t) \tag{38}$$

where $\hat{\mathbf{Q}}_t^{\pi_{\theta_a'}}(s_t, a_t)$ is the state–action value function

$$\hat{\mathbf{Q}}_t^{\pi_{\theta_a'}}(s_t, a_t) = \hat{\mathbb{E}}_{a_t \sim \pi_{\theta_a'}, s_t}[\mathbf{R}_t] \tag{39}$$

and $\hat{\mathbf{V}}_t^{\pi_{\theta_a'}}(s_t)$ is the state value function

$$\hat{\mathbf{V}}_t^{\pi_{\theta_a'}}(s_t) = \hat{\mathbb{E}}_{s_t}[\mathbf{R}_t]. \tag{40}$$

Furthermore, to mitigate unstable policy updates, we transform the loss function (37) into a clipped loss function

$$L^{\text{clip}}(\theta_a) = \hat{\mathbb{E}}_t\left[\min\left(\psi \cdot \hat{\mathbf{A}}_t^{\pi_{\theta_a'}}, \text{clip}(\psi, 1-\varepsilon, 1+\varepsilon) \cdot \hat{\mathbf{A}}_t^{\pi_{\theta_a'}}\right)\right] \tag{41}$$

where $\psi = (\pi_{\theta_a}(a_t|s_t)/\pi_{\theta_a'}(a_t|s_t))$ is the probability ratio. By controlling the probability ratio in the range $[1-\varepsilon, 1+\varepsilon]$, excessive policy updates can be prevented. Since θ_a is updated using gradient ascent, its parameters can be iteratively updated as follows:

$$\theta_a := \theta_a + \text{lr}_a \nabla\theta_a L^{\text{clip}}(\theta_a) \tag{42}$$

where lr_a is the learning rate of the actor.

Critic: Critic estimates state value $\hat{\mathbf{V}}_t^{\pi_{\theta_a}}(s_t)$ of each state s_t by policy π_{θ_a}. The loss function $L(\theta_c)$ of the critic network is defined as follows:

$$L(\theta_c) = \mathbb{E}\left[\left(\mathbf{R}_t - \hat{\mathbf{V}}_t^{\pi_{\theta_a}}(s_t)\right)^2\right]. \tag{43}$$

Meanwhile, the parameters θ_c of the critic network are updated through gradient descent with learning rate lr_c

$$\theta_c := \theta_c - \text{lr}_c \nabla\theta_c L(\theta_c). \tag{44}$$

During the interaction, the agent S first needs to acquire the current state s_t of the system and then selects an action a_t by actor network with policy $\pi_{\theta_a}(a_t|s_t)$. Subsequently, the state transitions to s_{t+1} and the agent S receives a reward r_{t+1}. After each round of interaction, the tuple $(s_t, a_t, r_{t+1}, s_{t+1})$ is stored in the experience replay buffer $\mathcal{D}_e$. During training, a mini-batch of experiences $\mathcal{B} = \{(s_t, a_t, r_{t+1}, s_{t+1})\}$ is sampled from $\mathcal{D}_e$ to update the network parameters.

The detailed training process of the PMSC-URA scheme is summarized in Algorithm 2.

Algorithm 2 Training the PMSC-URA Scheme

Input: Library dataset $\mathcal{D} = (X, Z)$, pretrained PMSC network $e(\cdot)$ and $d(\cdot)$, pretrained onboard task network $\phi(\cdot)$, training episode E_p, total time slots T, learning rate lr_a and lr_c of PPO network.

Output: The optimal PPO network parameters θ_a^*, θ_c^*.

```
Initialize the parameters θ_a, θ_c, the state s_t, and the experience replay memory D_e;
for episode = 1 : E_p do
    Reset s_t = s_0;
    for t = 1 : T do
        Sample image x_i(t) from D for UE_i as required;
        Extract the semantic information of image y_i(t) = e(x_i(t), φ(x_i(t)));
        Input s_t into the actor network and obtain a_t;
        S transmits y_i(t) to UE_i by taking a_t;
        The UEs feedback the ACK/NACK and then S obtains the reward r_{t+1} and s_{t+1};
        Store (s_t, a_t, r_{t+1}, s_{t+1}) in replay memory;
        Randomly select B from replay memory D_e;
        Estimate Â_t^{π_{θ_a'}}(s_t, a_t) by Eq. (38);
        Calculate L^clip(θ_a) according to Eq. (41);
        Update θ_a by ∇_θ L_clip(θ_a);
        Calculate L(θ_c) according to Eq. (43);
        Update θ_c by ∇_{θ_c} L(θ_c);
    end for
end for
return parameters θ_a^*, θ_c^*.
```

V. Simulations and Results Analysis

This section provides simulation results to validate the performance of the TUT framework. The simulations and subsequent analysis focus on two main aspects: 1) task-oriented image semantic coding performance and 2) UoI performance.

TABLE I
Object PM Values

Object	PM Value
plane, ship, harbor, helicopter	1
small-vehicle, large-vehicle, storage-tank, soccer-ball-field, roundabout, tennis-court, basketball-court, swimming-pool, baseball-diamond, bridge, ground-track-field	0.5
background	0

A. Semantic Coding Performance

Regarding the performance of the PMSC network, we conduct simulation tests to evaluate the performance of the PMSC network in two tasks relevant to satellite-integrated Internet.

1) Integration with onboard objection detection tasks for remote sensing images, where detection results are used to generate PMs.
2) Integration with onboard semantic segmentation tasks for remote sensing images, where different values are assigned to various semantic regions of the image.

1) Object Detection: The library dataset $\mathcal{D}$ utilized for object detection tasks is the DOTA-v1.0 dataset. We adopt Yolov5-obb for onboard object detection. Yolov5-obb performs objection detection on images, assigning higher values to regions containing objects of interest. Without loss of generality, we categorize the objects detected by Yolov5-obb and the background into distinct classes. The specific classifications and value assignments are provided in Table I. For comparison,

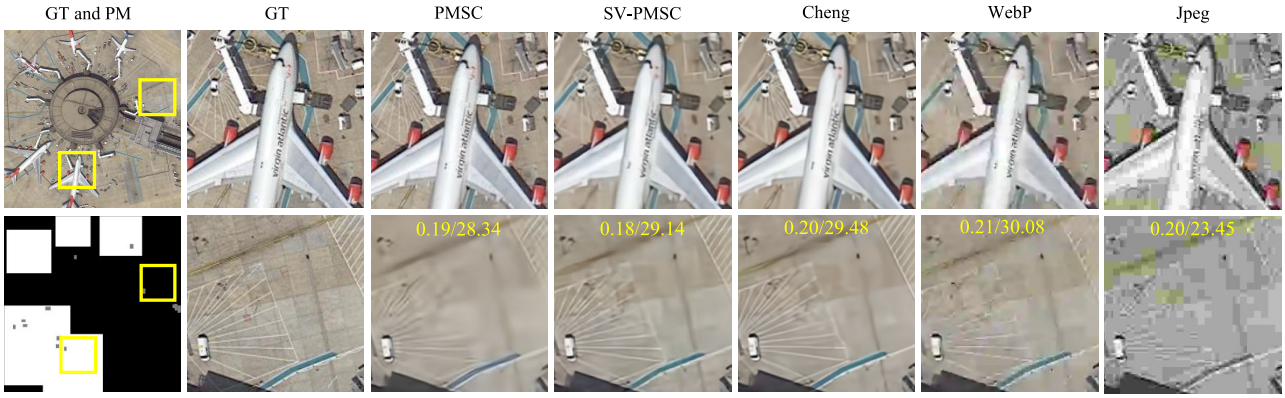
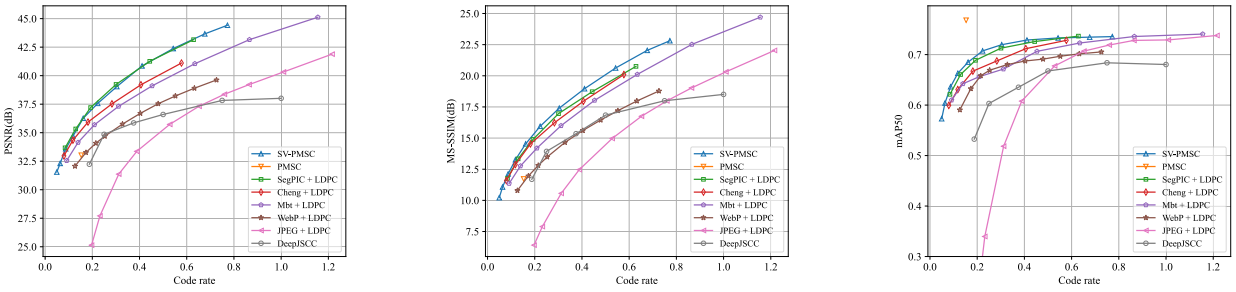


Fig. 3. Comparison between GT, PMSC, SV-PMSC, Cheng, and JPEG. The annotations in images indicate the CR/PSNR (dB) of the entire image.



(a) (b) (c)

Fig. 4. Performance comparison with the DOTA-v1.0 dataset. (a) PSNR. (b) MS-SSIM. (c) mAP50.

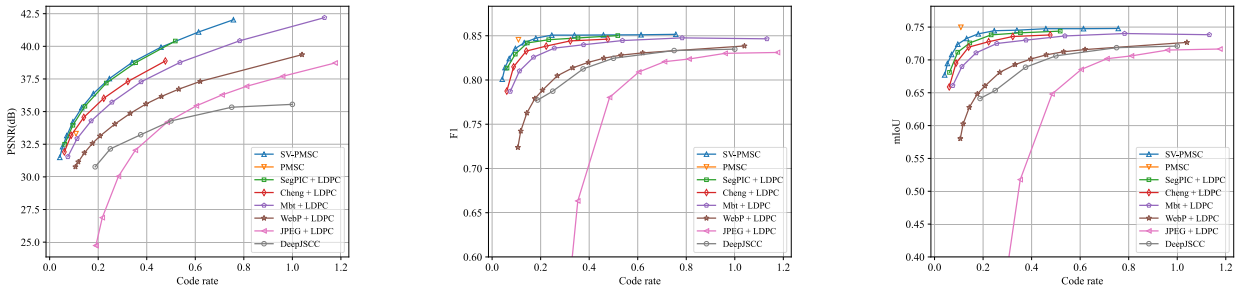
we use traditional image coding methods, including JPEG and WebP [59], and the DL-based methods Mbt [60], Cheng [61], and SegPIC [62] with low-density parity-check (LDPC) codes in 3/4 code rates, as well as DeepJSCC [10] as benchmark schemes.

2) Image Semantic Segmentation: Fig. 3 presents a comparison of PMSC network, which uses RoI-based PMs generated by Yolov5-obb according to Table I, SV-PMSC which uses single value PMs, and benchmark schemes on an image from the DOTA-v1.0 dataset at similar CR levels. The grayscale image in the bottom-left corner represents the PM generated by Yolov5-obb. Zooming in on specific regions reveals that in areas, where PM values are distributed differently, such as the high PM value assigned to the airplane and the low PM value assigned to the ground markings, the PMSC-RoI method successfully reflects the ROI information for the ability of SFT to generate element-wise affine using PM as external prior.

Fig. 4 illustrates the performance of PMSC, which using task-oriented RoI-based PMs generated by Yolov5-obb according to Table I. Note that the SV-PMSC is with artificially assigned single-value PMs (i.e., PM $=$ 0, 0.1, 0.2, $\ldots$, 1.0) to validate variable-rate capability, and benchmark schemes across CR levels at fixed SNR, where SNR $=$ 25 dB in shadowed-Rician channel with infrequent light shadowing, as measured by traditional metrics such as PSNR and MS-SSIM, and the semantic performance metric mAP for object detection. Notably, for PMSC, given an image dataset and a predefined task, each RoI in the dataset's images has a determined PM value, which corresponds to a fixed code rate. It can be observed in Fig. 4(a) and (b) that the SV-PMSC outperforms other schemes at various CR levels. This superior performance is attributed to the elimination of coding loss due to LDPC codes, leading to optimal scores for the traditional metrics PSNR and MS-SSIM, and the semantic performance metric mAP. In the PMSC integration with Yolov5-obb, Fig. 4(a) and (b) shows no significant advantage in the traditional metrics PSNR and MS-SSIM, as these metrics are not designed to assess and perceive image content. In contrast, the PMSC demonstrates remarkable superiority in mAP metric, due to the use of Yolov5-obb as the PM generator, as shown in Fig. 4(c). Notably, while Yolov5-obb achieves an mAP of only 0.745 on the original DOTA-v1.0 images, it achieves an mAP of 0.768 on the PMSC-RoI reconstructed images. This indicates that PMSC not only preserves essential semantic information at higher CR but also enhances it.

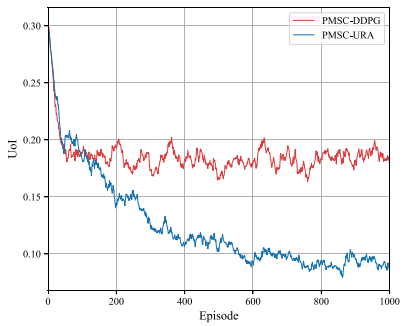
For the semantic segmentation task of remote sensing images, we conduct tests on the Potsdam dataset, using the Unetformer [63] as the onboard task model. The outputs of the Unetformer serve as the PMs. The Potsdam dataset contains six semantic labels: impervious surfaces, trees, cars, buildings, low vegetation, and clutter/background, with the assigned corresponding quality values set as 0, 0.25, 0.5, 0.5, 1, and 0.25, respectively.

Fig. 5 illustrates the performance of PMSC, SV-PMSC, and benchmark schemes on the Potsdam dataset across various CR levels. It can be observed in Fig. 5(a) that while PMSC does not show a significant advantage in traditional metrics like the PSNR, it excels in semantic performance metrics such as $F1$ and mIoU as shown in Fig. 5(b) and (c), respectively, which evaluate image semantic segmentation performance.

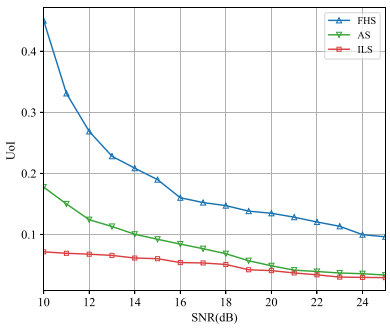


(a) (b) (c)

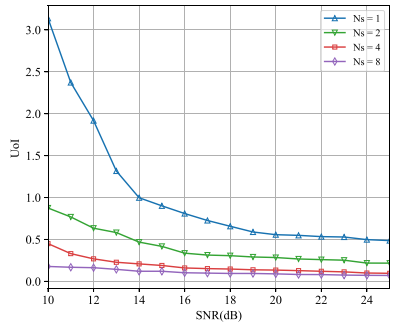
Fig. 5. Performance comparison with the Potsdam dataset. (a) PSNR. (b) $F1$. (c) mIoU.



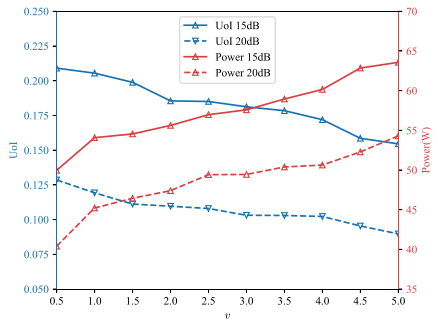
(a) $\bar{\Delta}_{\text{UoI}}$ changes with training episode.



(b) $\bar{\Delta}_{\text{UoI}}$ under different channel fading types.



(c) $\bar{\Delta}_{\text{UoI}}$ with different N_s under FHS.



(d) Tradeoff between $\bar{\Delta}_{\text{UoI}}$ and $\bar{\mathbf{P}}$.

Fig. 6. (a) $\bar{\Delta}_{\text{UoI}}$ changes with training episode, where $K = 4$, $p_r = 0.5$, $v = 1.5$, $N_s = 4$, and FHS. (b) and (c) $\bar{\Delta}_{\text{UoI}}$ versus SNR of the PMSC-URA scheme, where $K = 4$, $p_r = 0.5$, $v = 1.5$. (b) $N_s = 4$. (d) Tradeoff between $\bar{\Delta}_{\text{UoI}}$ and $\bar{\mathbf{P}}$ of the PMSC-URA scheme versus v in different SNRs, where $K = 4$, $p_r = 0.5$, and $N_s = 4$.

TABLE II
PARAMETERS AND ENCODING TIMES

Method	Parameters (MB)	Encoding time (s)
PMSC	317 (single model for all CRs)	0.43
Cheng [61]	54.3-121 (for each CR)	0.26
SegPIC [62]	319 (for each CR)	0.32
JPEG	\	0.09

TABLE III
SIMULATION SETUP

Parameter	Value	Parameter	Value
P_{mean}	100 W	Clip function parameter ε	0.2
$P_{\max}$	200 W	Training episodes E_p	1000
Time slot τ	5 ms	Time slots per episode	200
Discount factor γ	0.9	State remain probability p_r	0.5

This trend is consistent with PMSC's performance in the object detection task.

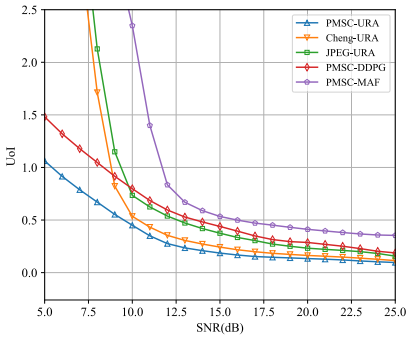
3) Model Parameters and Encoding Times: As evidenced in Table II, the proposed PMSC is capable of achieving a wide range of CRs via a single model, contrasting with other DL approaches that necessitate distinct models per CR. This unified architecture eliminates redundant parameters inherent in maintaining multiple CR-specific models while enhancing deployment scalability in resource-limited scenarios.

B. UoI Performance

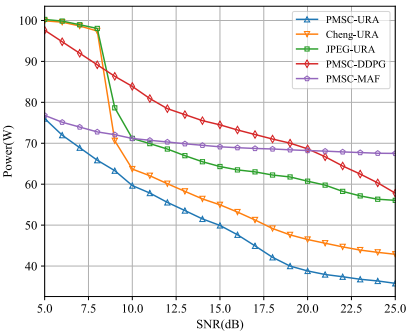
We conduct simulations to assess the performance of the proposed PMSC-URA scheme. The simulation is based on the first scenario, which involves the integration with object detection. The dataset and object detection model remain consistent with previous simulations. Table III lists the key simulation parameters for training in the TUT framework,

TABLE IV
SHADOWED-RICIAN FADING PARAMETERS

Fading type	b_i	Ω_i	m_i
Infrequent light shadowing (ILS)	0.158	1.29	20
Average shadowing (AS)	0.126	0.835	11
Frequent heavy shadowing (FHS)	0.063	0.000897	1



(a)

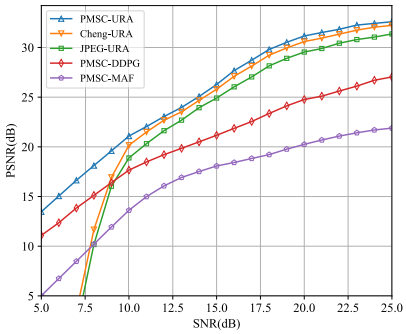


(b)

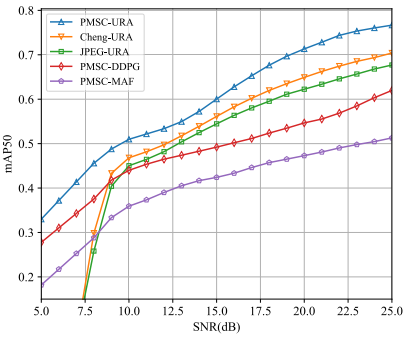
Fig. 7. Performance of PMSC-URA and benchmark schemes, where $K = 4$, $p_r = 0.5$, $v = 1.5$, and $N_s = 4$. (a) $\bar{\Delta}_{\text{UoI}}$ versus SNR with different schemes. (b) $\bar{\mathbf{P}}$ versus SNR with different schemes.

while the shadowed-Rician fading parameters for different types of channels are presented in Table IV.

We consider a scenario with four UEs ($K = 4$). Fig. 6(a) demonstrates the convergence of our PMSC-URA scheme along with training, which is faster than the DRL-based PMSC-DDPG scheme [46]. Fig. 6(b) illustrates the impact of various shadowed-Rician fading types on the long-term average UoI in the PMSC-URA scheme, where the UoI decreases as the SNR increases. The influence of SNR on UoI is particularly pronounced under more severe fading conditions, especially when the SNR is less than 16 dB. Furthermore, the primary differences in UoI between FHS, AS, and ILS stem from the limited power of S. To validate the effectiveness of the PMSC-URA scheme, we employ FHS fading in subsequent simulations for comparative analysis. Fig. 6(c) demonstrates that UoI improves with an increasing number of transmitting antennas N_s under FHS, following the same reason related to the SNR. Fig. 6(d) illustrates the tradeoff between $\bar{\Delta}$UoI and average power consumption $\bar{\mathbf{P}}$ by varying the weight coefficient v. As shown in Fig. 6(d), increasing v leads to a reduction in $\bar{\Delta}$UoI, but this comes at



(a)



(b)

Fig. 8. Performance of PMSC-URA and benchmark schemes, where $K = 4$, $p_r = 0.5$, $v = 1.5$, and $N_s = 4$. (a) PSNR versus SNR with different schemes. (b) mAP versus SNR with different schemes.

the expense of increased $\bar{\mathbf{P}}$. In the Lyapunov DPP term, the primary function of v is to balance the minimization of UoI with the stability of the TUT framework. These results further confirm the intricate relationship between stability and power consumption.

Next, we compare the PMSC-URA scheme with other benchmark schemes. For the NOMA resource allocation, two schemes are considered: 1) MAF scheme [47], which prioritizes the allocation of maximum available power to the UE$_i$ that with the highest UoI and 2) the DRL-based DDPG scheme [46]. In the specific simulations, we combine various image coding methods like Cheng et al. [61], and JPEG with resource allocation policies for comparison. It is essential to highlight that, to guarantee that UEs can effectively perform their tasks after receiving the information, we impose a constraint on the image code rate by setting a minimum semantic performance threshold. Specifically, for the object detection task, we require that the reconstructed images achieve a minimum mAP of 0.7 at SNR = 25dB.

As illustrated in Fig. 7(a), the PMSC-URA scheme maintains superior UoI performance over varying SNR levels. The JSCC-based architecture, combined with PMSC's efficient semantic feature extraction, leads to a marked enhancement in UoI compared to Cheng-URA and JPEG-URA schemes, both suffering from the characteristic cliff effect under low SNR conditions attributed to their disjoint source-channel coding architectures. Even when employing PMSC for image coding, the PMSC-URA scheme outperforms both the PMSC-MAF and PMSC-DDPG schemes, due to the high training effi-

ciency and adaptability of the URA algorithm, underscoring its effectiveness in complex and dynamic satellite-terrestrial environments. Fig. 7(b) depicts the power consumption of the PMSC-URA scheme and other schemes. Notably, all DRL-based approaches demonstrate environmental awareness, with power consumption decreasing as SNR increases. In contrast, the traditional resource allocation scheme, MAF, maintains nearly constant power consumption regardless of SNR. Importantly, the PMSC-URA scheme achieves optimal UoI performance while maintaining lower power consumption across different SNR levels.

Fig. 8(a) and (b) shows the PSNR and mAP of the reconstructed images at the UEs, respectively. The simulation results demonstrate that the PMSC-URA scheme not only provides UEs with the highest image quality but also enables efficient task completion. In Fig. 8(b), the higher mAP indicates that the UEs received more relevant information, facilitating more effective task performance. When combined with the UoI trend in Fig. 7(a), these findings further validate that the UoI metric reflects the efficiency of task performance.

VI. Conclusion

In this article, we proposed a task-oriented semantic coding and utility-optimal transmission framework, named TUT, for satellite-integrated Internet. A task-aware metric, UoI, which captures freshness, task updates, and task completion of UEs was proposed to comprehensively evaluate system performance. We designed the PMSC network, which supports flexible semantic coding across multiple tasks with adjustable code rates through a unified structure. By dynamically adjusting the numerical distribution of PM, we achieved optimized semantic coding tailored to UoI. We then formulated a long-term average UoI minimization problem under multiple constraints in the TUT framework. Using the Lyapunov optimization framework, we decomposed the problem and addressed it using the PMSC-URA algorithm. Simulation results showed that the PMSC-URA scheme achieved the lowest power consumption and UoI, ensuring timely and reliable image transmission as well as task completion of UEs. The proposed PMSC-URA scheme can be extended to scenarios involving multiple modalities and tasks in future work.

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